



CSC205 ARTIFICIAL INTELLIGENCE

Lecture # : 03 & 4

Topic: Types Of AI Agents

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Learning Outcomes



Understand the **concept and types of AI agents**.



Differentiate between **reflex, model-based, goal-based, utility-based, and learning agents**.



Identify **real-world applications** of different AI agents.

AI System

AI is defined as study of relational agents.

Agent : Responsible for any workout obtain from the systems

AI Systems composed of two thing:

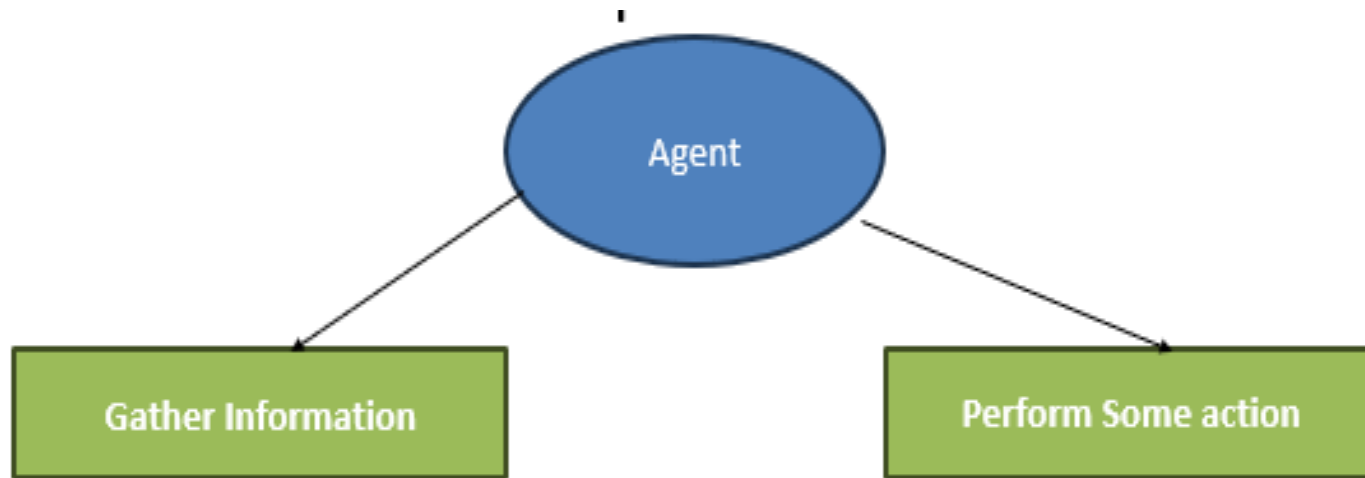
1. **Agents**

2. **Environments**



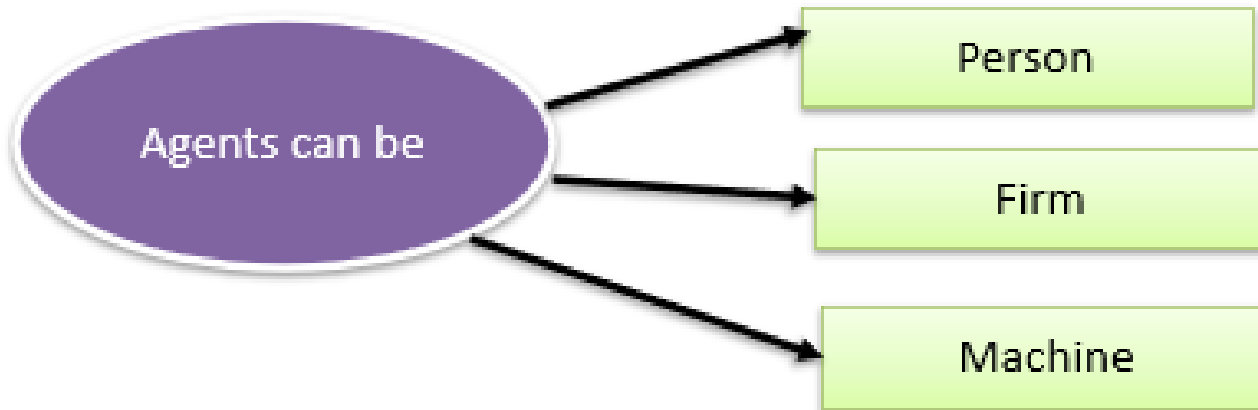
Agents

Agent can perceive some information from the environment or can perform some actions.



Agents

Agents are Rational agents:
that means they can make decision



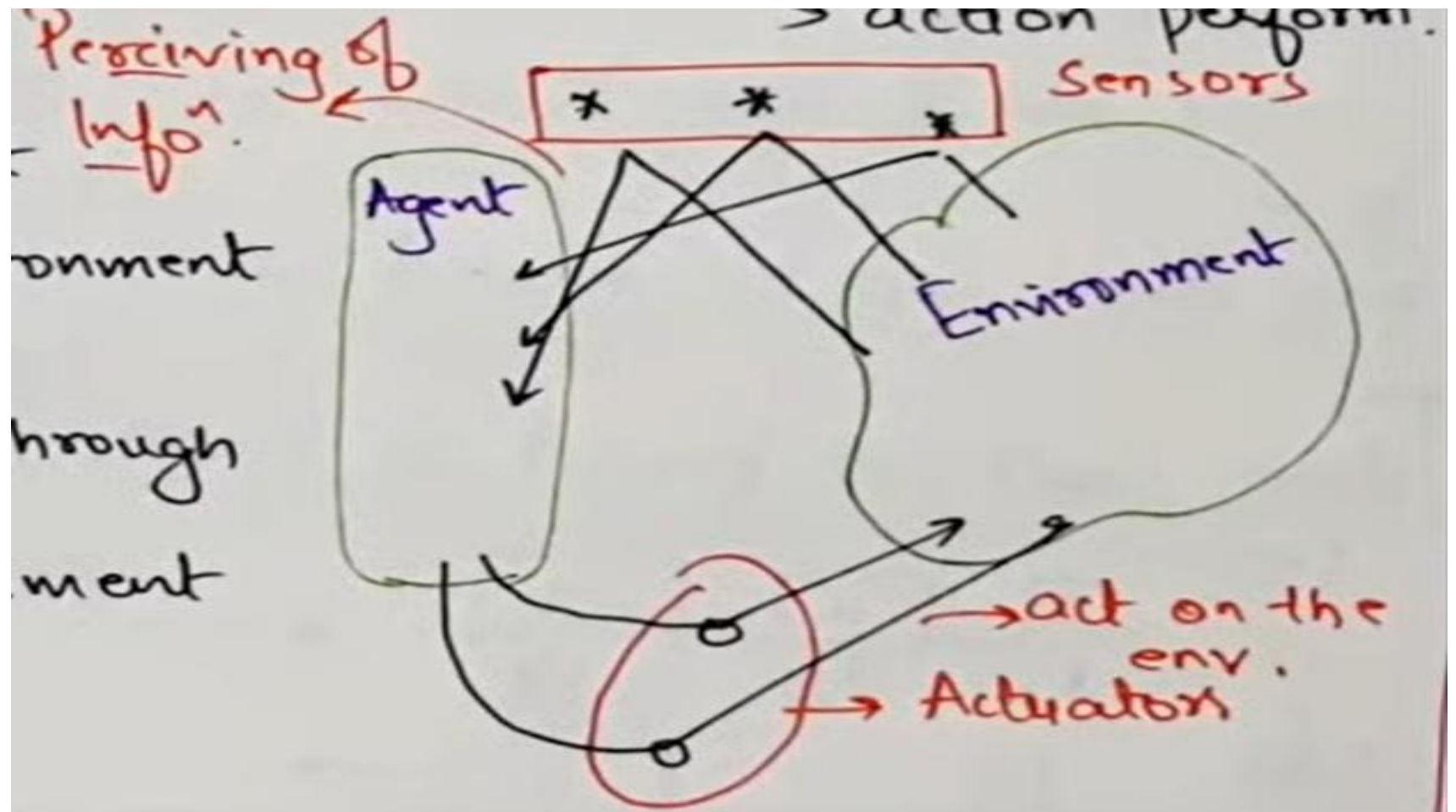
AI Agents

Agents is anything like:

Perceiving its environment through **sensors**
and

acting upon that environment through **actuator**.





- An **AI agent** is responsible for the actions and outputs produced by a system based on the information it receives from its environment.
- An agent **perceives information from its environment** and takes appropriate actions based on that information.



- **Artificial Intelligence can be understood as the study of intelligent agents** that perceive their environment and act rationally.
- A **rational agent** makes decisions that are expected to achieve the best possible outcome based on the information available to it.
- An agent can be a **person, organization, software system, robot, or machine.**



Real world example

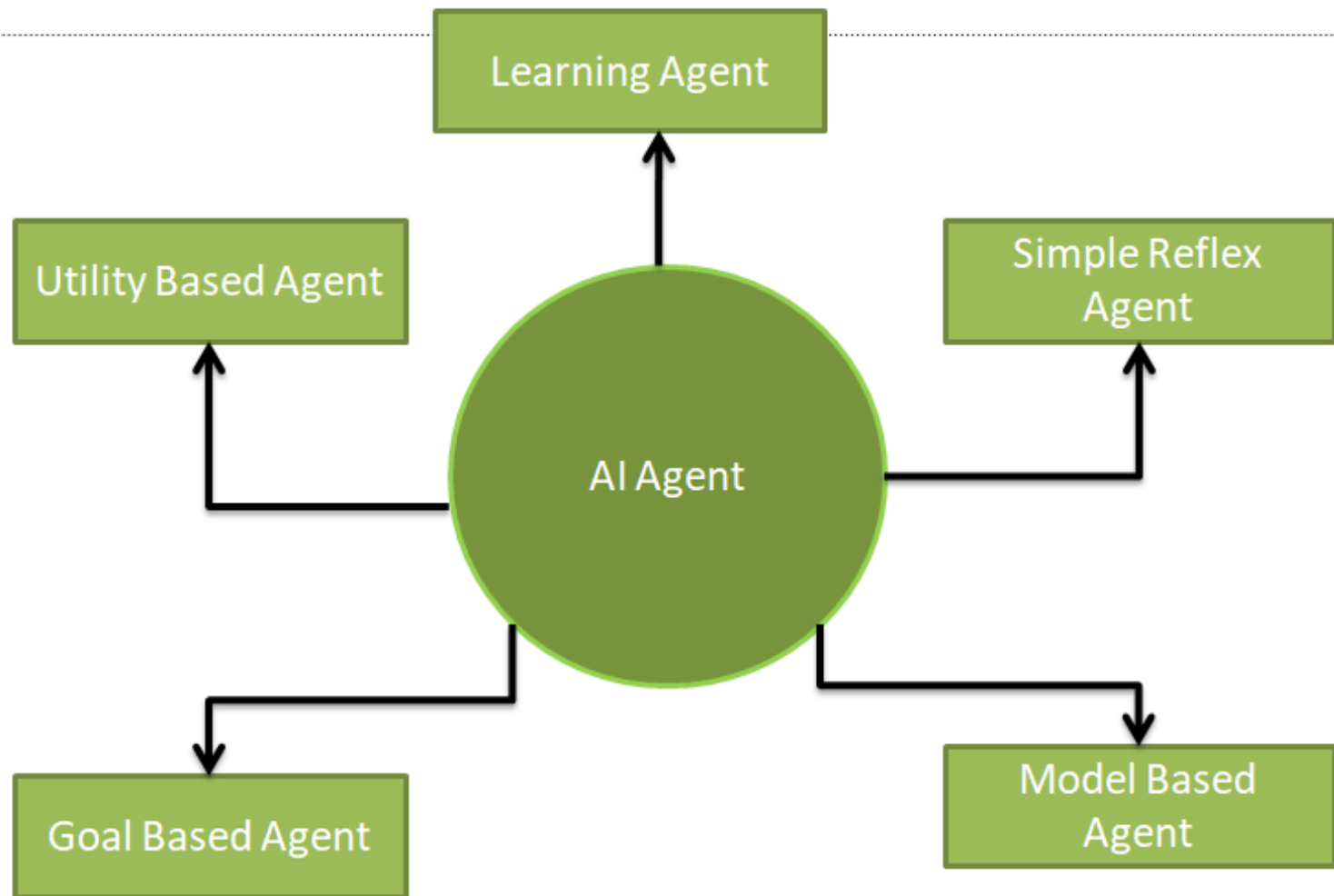
- Suppose you set your air conditioner to **25°C**.
- The **temperature sensor** continuously measures the room temperature.
- If the room temperature is **above 25°C**, the AC continues cooling.
- When the temperature reaches **25°C**, the system detects this through its sensor.
- The controller then sends a command to the **actuator**, which turns the cooling system **OFF**.
- If the temperature rises again, the sensor detects the change and the AC turns **ON** again.



What is an AI Agent?

- An AI agent is a system that:
 - Perceive the environment
 - Process information
 - Takes actions (takes actions through actuator)
 - Actuator are devices.
 - Agent = Architecture + program
 - Objective is to maximize performance measure.





Simple reflex Agent

- It followed pre define rules to make decisions.
- It only response to the current situation without considering the past and future ramification.
- Works well in fully observable environments.
- Fails when information is incomplete.
- If something happen agent gonna take action



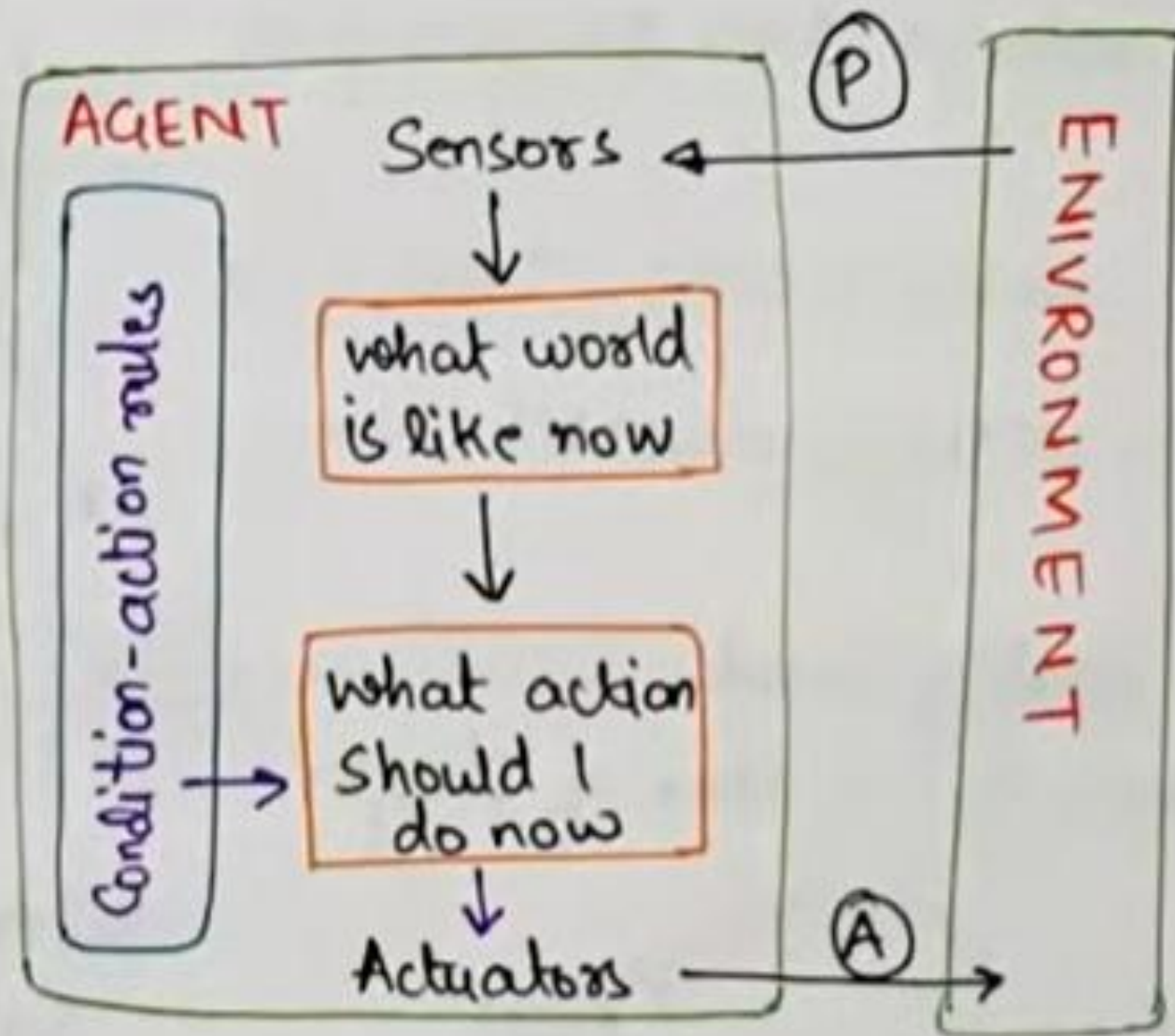
Simplex Agent

- Works only on current situation/ perception and ignores the history of previous state.

Condition Action Rule

Limitations:

- Very limited Intelligence.
- No knowledge about non perceptual parts of state.
- Can go into infinite loop.



REAL WORLD EXAMPLE

Traffic Light Controller

A traffic light follows simple rules:

- If light is GREEN – allow vehicles to go.
- If light is RED – stop vehicles.
- If light is YELLOW – get ready to stop.

It only looks at the current signal and does not consider past traffic or future congestion.



Like if there's smoke then beep the alarm

WHEN IT FAILS – REAL WORLD EXAMPLE

Smart Home Motion Light

If a motion sensor light only uses the current input, it may turn ON when a small animal passes by, even if no one is around. It doesn't remember past patterns or understand context.



- An automatic door can also behave like a Simple Reflex Agent:
- **IF** motion is detected → **open the door**
IF no motion is detected → **close the door**
- It simply responds to the **current sensor input**.



Why does it fail with incomplete information?

- Imagine a motion sensor detects a **cat** instead of a person. If the system only knows:
- “Motion detected → Open door”
- it will open the door because it **doesn't understand the context** or remember previous situations.



Model based Agents

- Remembered what happened in the past and makes better decisions based on its memory.
- Maintains an internal state .
- $\text{Current state} = f(\text{previous state, percept})$
- Handles partially observable environments.
- More intelligent than reflex agents.



Model Based Agents

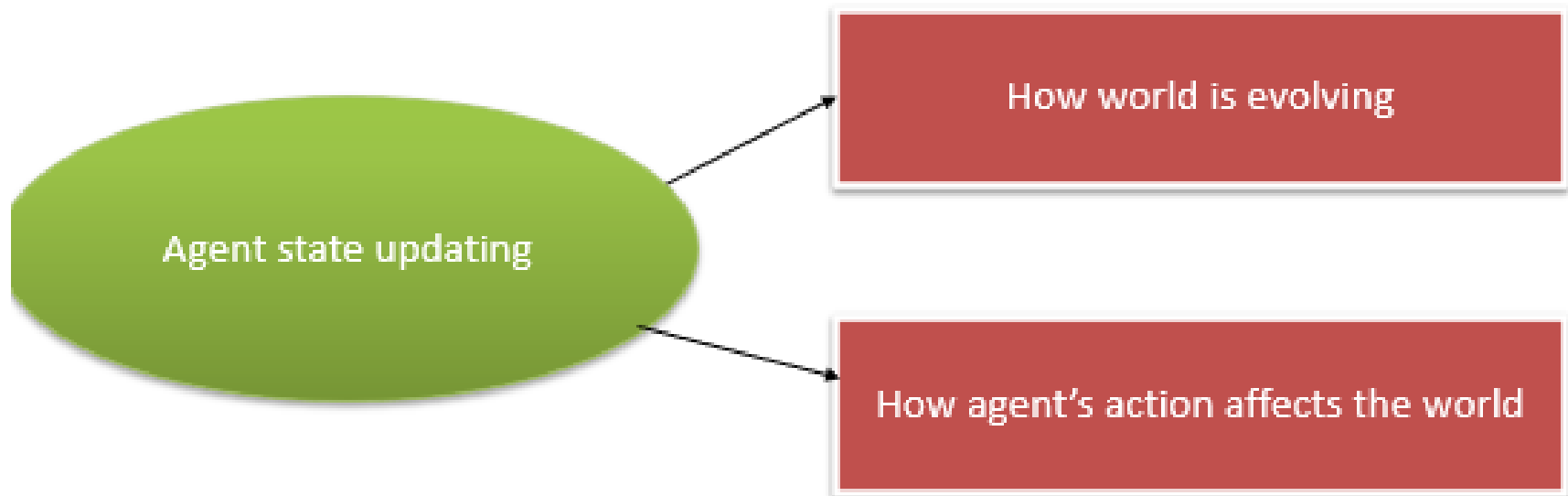
Works by finding the rules whose condition matches current situations.

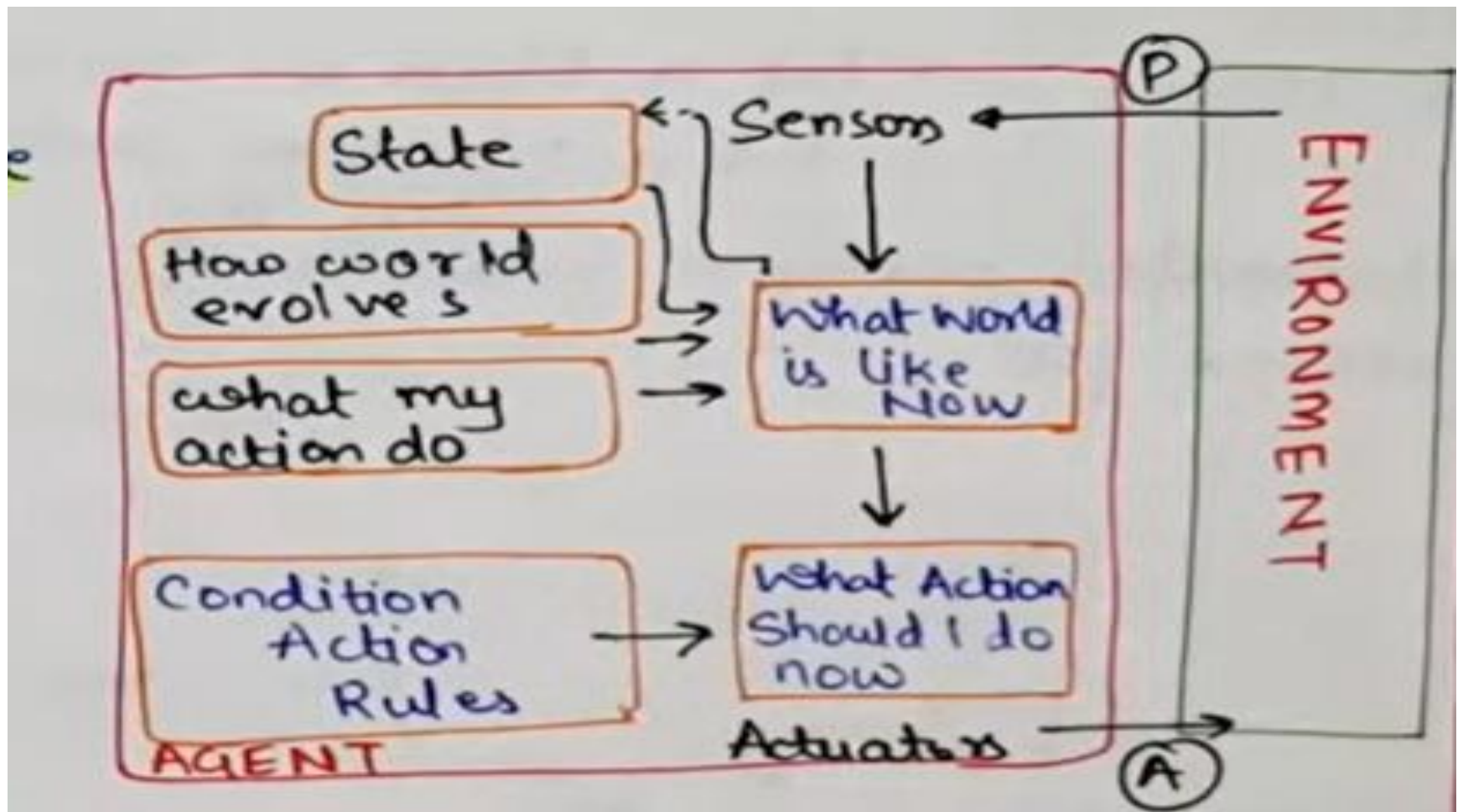
Can work in partially observable environment and track situation.

Agents keep track of an internal state which is adjusted by each percept and that depends on the percept history.

Model: How things happen in the world.







Robot Vacuum Cleaner

Detects obstacles around it.

Remembers areas it has already cleaned.

Uses its internal map to decide where to go next.

Avoids repeatedly cleaning the same area.

**Environment → Sensors → Internal Model →
Decision → Action**

Key Point

👉 *“I remember what happened, so I can decide better.”*



A self-driving car

1. Sensors perceive the environment

Cameras, LiDAR, and radar detect cars, pedestrians, traffic lights, and road conditions.

2. It remembers previous information

The car remembers what it detected a moment ago—for example, a pedestrian was walking toward the road.



3. It maintains an internal state

It combines its **previous state** with the **current perception** to understand what is happening.

Current State = f(Previous State, Current Percept)

4. It handles incomplete information

Suppose a car temporarily blocks the view of a pedestrian. The agent can use its previous information to estimate what may still be happening.



5. It makes a better decision

Instead of simply reacting to what it sees right now, it can **brake, slow down, or change direction** based on the current situation + previous information.





Goal Based Agent

- An agent that takes actions to achieve a **specific goal**.
- It considers **different possible actions** before making a decision.
- It asks: “**What should I do to reach my goal?**”
- More flexible than a simple reflex agent.



Goal Based Agent

- Goal Based agents are agents that use information from their environment to achieve specific goals.
- They employ the search algorithms to find the most efficient path towards their objectives within a given environment.



Google Maps

Goal: Reach a destination.

It considers different routes.

It selects a route that can help you reach the destination.

If the road is blocked, it can find another route.

Simple Flow

**Current State → Possible Actions → Goal → Best Action
→ Result**

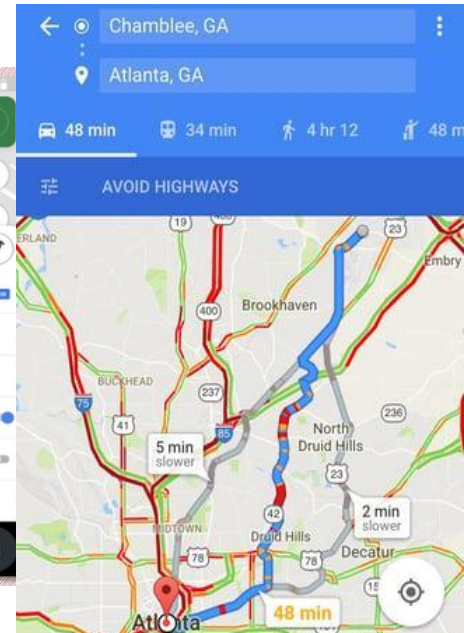
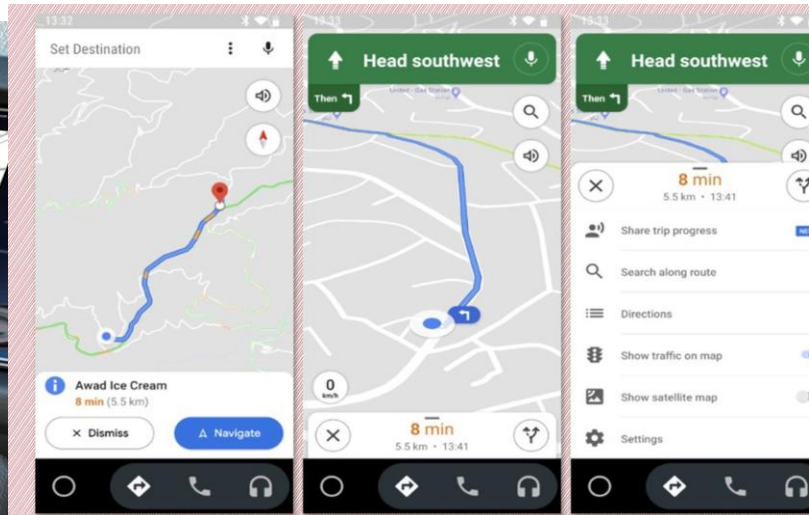
Key Point

👉 *“I choose actions that help me achieve my goal.”*



Google Map

- A **Goal-Based Agent** can be understood through a navigation system such as Google Maps:





Goal: Reach a specific destination.



Environment Information: Current location, roads, traffic, road closures, and distance.



Search: It evaluates different possible routes



Decision: It selects the most suitable/efficient route.



Adaptation: If traffic or road conditions change, it can find another route.



Utility Based Agents

- An agent that chooses the action with the **best overall outcome**.
- It uses a **utility function** to evaluate different choices.
- It considers **multiple factors**, not just one goal.
- It tries to maximize **benefit, satisfaction, or performance**.



Utility Based Agents

- Utility based agents are AI agents that make decision based on maximizing a utility function or a value.
- They choose the action with the highest expected utility , which measures how good the outcome is.
- Take it like a recommendation systems



Ride-Hailing App

When choosing a ride, the system may consider:



Price



Waiting time



Distance



Driver rating

It selects the option that provides the **best overall choice**.

Simple Flow

Possible Actions → Evaluate Utility → Select Best Option
→ Action

Key Point



“Which choice gives me the best overall result?”



Utility Based Systems

A utility-based agent is an AI system that selects actions based on a utility function, which quantifies how desirable a particular outcome is. Unlike goal-based agents that operate with predefined objectives, utility-based agents evaluate multiple options and choose the one that maximizes overall benefit or utility.



This makes them more flexible and capable of handling complex decision-making scenarios. They focus on improving decision-making by considering various factors such as cost, time, and risk to select a particular action that maximises overall utility.



How do utility-based agents work?

- **Perception:** The agent perceives its environment through sensors and collects relevant data to predict the potential actions.
- **Utility Function:** The system uses a mathematical function to assign a utility value to each potential action. The function may consider multiple factors like probability, cost, and expected rewards.



- **Decision-Making:** The agent evaluates different actions using the utility function and selects the one that maximizes utility.
- **Execution and Observe:** The chosen action is executed in the environment. It also take feedback and observes new state to update knowledge and refine future decisions.



- **Learning and Adaptation:** Some utility-based agents employ machine learning to improve their utility function over time based on past experiences and improve decisions and performance over time.



Role of utility based Agents

Autonomous Vehicles

Self-driving cars rely on utility-based decision-making to navigate safely



01

Recommendation Systems

Recommend products and services based on users' search engine activity.



02



03

Finance

Algorithmic trading systems use utility-based decision-making to optimize stock investments.



04

Healthcare

Support treatment decisions using patient history, symptoms, and success probabilities.



Role of utility based Agents

- **Autonomous Vehicles:** Self-driving cars rely on utility-based decision-making to navigate safely. They evaluate multiple factors, such as traffic conditions, pedestrian movement, fuel efficiency, and weather, to determine the best route.



- **Recommendation Systems:** Utility-based agents are used to recommend products and services to potential customers on the basis of their search on search engines, scrolling activity, and buying patterns.



- **Finance:** Algorithmic trading systems employ utility-based decision-making to optimize stock market investments. These agents assess market trends, risk levels, and potential returns before executing trades.



- **Logistics:** Utility-based agents can be useful by automating tasks like route optimization, warehouse operations, and inventory management.
- **Healthcare:** It can be useful in assisting doctors and other healthcare professionals in treatments based on patient history, symptoms, and success probabilities.



Advantages

- **Optimized Decision-Making:**
- **Flexibility:**
- **Scalability**
- **Risk Management:**
- **Enhanced Personalization:**



Learning based Agents

- An agent that **learns from experience and data**.
- It improves its performance over time.
- It can adapt when the environment changes.
- It does not always depend on fixed rules.
- **Real-World Example** 



Learning Agent

A **Learning Agent** is an AI agent that **improves its performance over time by learning from experience and feedback.**

- Improve performance over time.

Core concepts:

- Performance element
- Learning element
- Critic feedback
- Problem generator.



Netflix Recommendation System

Observes what you watch.

Learns your preferences.

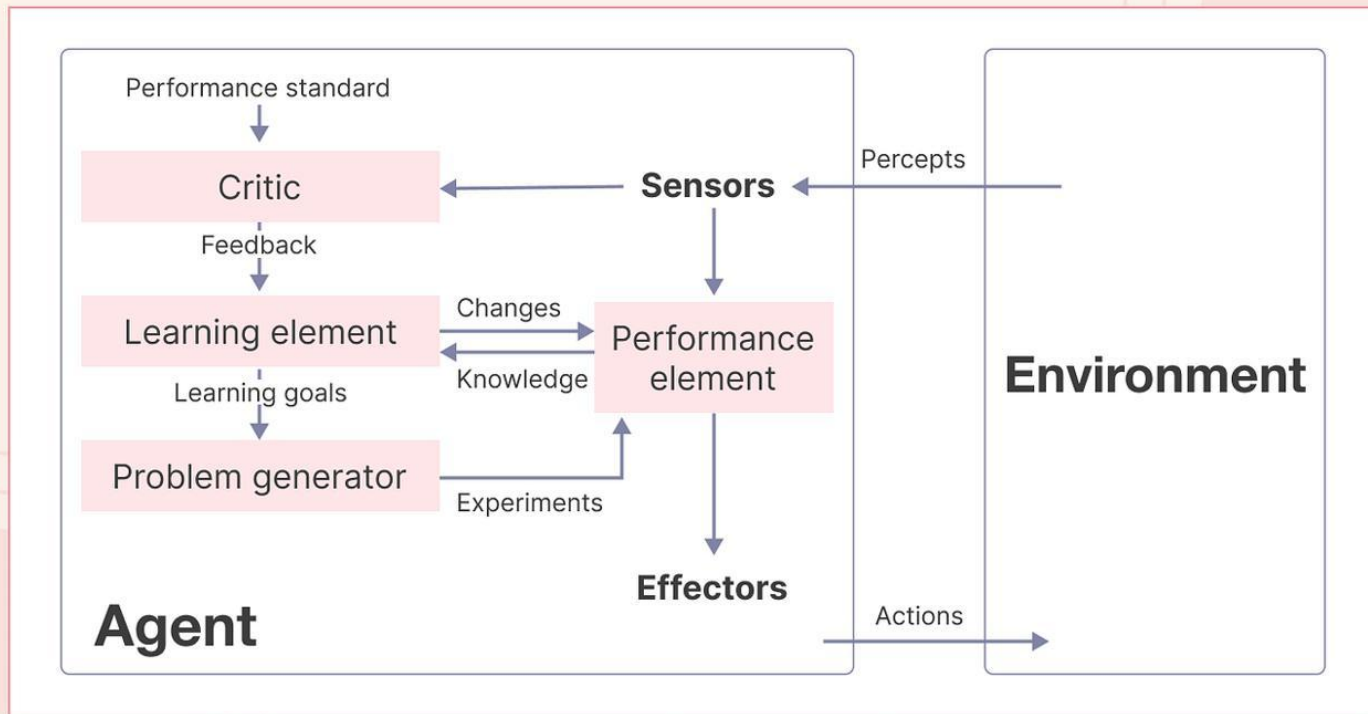
Finds patterns in your viewing behavior.

Recommends movies and shows you may like.



Learning Agent

Learning Agent



Real world example

- A YouTube recommendation system can be understood as a **Learning Agent**.
- **Performance Element:** Recommends videos to the user.
- **Learning Element:** Learns from the user's viewing behavior.
- **Critic:** Evaluates whether recommendations were successful—for example, whether the user watched, liked, skipped, or disliked a video.
- **Problem Generator:** Suggests new or different types of videos to explore and learn what the user might like.



Type of Agent	Simple Meaning	Real-World Example
1. Simple Reflex Agent	Acts only on the current situation using predefined rules. It does not remember the past.	 Automatic door: If a person is detected → open the door. If no person → close it.
2. Model-Based Agent	Uses the current situation + information about what happened before to understand the environment.	 Robot vacuum: Remembers which areas it has already cleaned and uses this information to navigate.
3. Goal-Based Agent	Takes actions to achieve a specific goal . It considers different possible actions before deciding.	 Google Maps: Goal = reach a destination. It selects a route that can take you there.
4. Utility-Based Agent	Chooses the action that provides the best overall result or highest satisfaction based on several factors.	 Ride-hailing app: Selects a ride considering price, waiting time, driver availability, and distance.
5. Learning-Based Agent	Learns from experience/data and improves its decisions over time.	 Netflix recommendation system: Learns from what you watch and improves future recommendations.

Activity 1: Match the Following

Agent Type	Example
1. Simple Reflex Agent	A. Netflix recommendations
2. Model-Based Agent	B. Google Maps
3. Goal-Based Agent	C. Automatic door
4. Utility-Based Agent	D. Robot vacuum
5. Learning-Based Agent	E. Ride-hailing app



Activity 2: Think & Answer



A self-driving car needs to reach a destination while considering **safety, fuel consumption, travel time, and traffic.**

Which type of agent is most suitable? **Why?**



Activity 3: One-Minute Quiz



- Q1. Which agent reacts only to the current situation?
- Q2. Which agent uses an internal model of the environment?
- Q3. Which agent works toward a specific goal?
- Q4. Which agent chooses the best overall outcome?
- Q5. Which agent improves through experience?

Answer

- Simple Reflex
- Model-Based
- Goal-Based
- Utility-Based
- Learning-Based

